

MediVerify Blockchain Explanation

Blockchain Kya Hoti Hai?

Sabse pehle blockchain ka simple matlab samajhte hain.

Agar hum normal database (MySQL ya Prisma) use karte hain, to admin ya jis ke paas access ho woh data ko kabhi bhi edit ya delete kar sakta hai.

Lekin blockchain ek aisa digital ledger (register) hai jahan ek dafa data likh diya jaye to usay na edit kiya ja sakta hai aur na hi delete kiya ja sakta hai. Har transaction permanently save ho jati hai aur uski history hamesha ke liye record rehti hai.

Isi wajah se blockchain ko tamper-proof aur immutable technology kaha jata hai.

Blockchain Ka Asal Maqsad Kya Hai?

Blockchain ka asal maqsad trust create karna hai.

Iska matlab ye hai ke agar koi medicine register hui hai ya verify hui hai to uska record kabhi change nahi ho sakta.

Is se fake medicines, duplicate records aur fraud ko rokna bohot asaan ho jata hai.

Mere Project MediVerify Mein Blockchain Ka Kya Role Hai?

Mere project ka objective fake medicines ko detect karna aur medicines ki authenticity verify karna hai.

Iske liye maine blockchain ka complete architecture design kiya hua hai.

Project ke andar blockchain ke liye Smart Contract bhi bana hua hai aur backend service bhi tayyar hai jo application ko blockchain se connect karegi.

Yani blockchain ka backend architecture already project me mojud hai.

Smart Contract Kya Kaam Karega?

Smart Contract blockchain ka main program hota hai.

Mere project me Smart Contract ye kaam karega:

- Medicine batch ko blockchain par register karega.
- Har medicine pill ka unique QR blockchain par save karega.
- Manufacturer ki information verify karega.
- Har verification ka permanent record rakhega.
- Agar koi QR code duplicate scan ho to uski detection karega.
- Har transaction ka immutable history maintain karega.

Ye sab kaam blockchain ke through honge.

Blockchain Service Kya Karegi?

Application aur blockchain directly baat nahi karte.

Isliye backend me Blockchain Service banai gayi hai.

Ye service:

- Smart Contract ko call karegi.
- Medicine batch blockchain par bhejegi.
- Verification transaction send karegi.
- Transaction Hash receive karegi.
- Aur us hash ko database me store karegi.

Ye application aur blockchain ke darmiyan bridge ka kaam karti hai.

Database Aur Blockchain Dono Kyun Zaroori Hain?

Application ka normal data database me hi rahega.

Jaise:

- Users
- Medicines
- Dashboard
- Inventory
- Reports

- Notifications

Lekin authenticity aur verification se related permanent records blockchain par store honge.

Is tarah system fast bhi rahega aur secure bhi.

Blockchain Workflow Kese Chalega?

Sabse pehle manufacturer medicine batch register karega.

↓

Batch database me save hoga.

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Blockchain Service us batch ko Smart Contract ke through Polygon Blockchain par bhejegi.

↓

Blockchain transaction complete hogi.

↓

Ek Transaction Hash generate hoga.

↓

Ye Transaction Hash database me save ho jayega.

↓

Jab bhi koi patient ya pharmacy QR scan karegi to verification event bhi blockchain par permanently record ho jayega.

Is tarah har medicine ki complete history available hogi.

Abhi Blockchain Fully Enable Kyun Nahi Ki?

Ye sabse important point hai.

Mera project abhi development stage me hai.

Testing ke dauran batches baar baar create hote hain.

QR Codes regenerate hote hain.

Data edit bhi hota hai.

Agar is stage par blockchain fully enable kar di jaye to jo bhi testing data blockchain par chala jayega woh kabhi delete nahi ho sakega.

Blockchain ki sabse badi property hi ye hai ke uska data permanent hota hai.

Isi wajah se maine abhi blockchain ka architecture complete kiya hai lekin uski live deployment intentionally disable rakhi hai.

Ye decision development ko asaan banane ke liye liya gaya hai.

Jab Project Live Hoga To Kya Hoga?

Jab ye project production me deploy hoga ya kisi company ko provide kiya jayega tab:

- Smart Contract Polygon Network par deploy kiya jayega.
- Wallet configure kiya jayega.
- Contract Address add kiya jayega.
- Blockchain Signer activate kiya jayega.
- Har medicine batch blockchain par permanently register hoga.
- Har QR verification blockchain par record hogi.
- Har transaction ka live Transaction Hash generate hoga.
- PolygonScan ke through koi bhi us transaction ko verify kar sakega.

Us waqt blockchain fully functional hogi.

Important Point

Agar aap poochein ke blockchain abhi live kyun nahi hai to uska jawab ye hai:

"Blockchain ka architecture, Smart Contract aur backend integration project me already tayyar hai. Lekin kyun ke project abhi development aur testing phase me hai, isliye real blockchain transactions intentionally disable rakhi gayi hain. Blockchain ka data permanent hota hai aur usay edit ya delete nahi kiya ja sakta. Isi wajah se production deployment ke waqt blockchain ko activate kiya jayega taake sirf final aur authentic medicine records hi permanently blockchain par store hon."

Conclusion

MediVerify me blockchain sirf ek concept nahi hai.

Iska Smart Contract, backend service aur complete architecture already develop kiya ja chuka hai.

Ab sirf production deployment, wallet configuration aur Polygon network par Smart Contract deploy karna baqi hai.

Uske baad system har medicine ki tamper-proof aur immutable verification history blockchain par permanently maintain karega.